Install Docker

On server 1

```
sudo apt install -y docker.io
```

Start Docker

```
sudo systemctl enable docker
sudo systemctl start docker
```

Check Docker status

```
sudo systemctl status docker
```

Check Docker version

```
docker --version
```

Run Docker without sudo

```
usermod -aG docker $USER
```

then logout and login, or reboot

Check Docker

```
Groups
```

```
verra@server1:~$ sudo apt install -y docker.io
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base pigz runc ubuntu-fan
Suggested packages:
  ifupdown aufs-tools cgroupfs-mount | cgroup-lite debootstrap docker-buildx docker-compose-v2
  docker-doc rinse zfs-fuse | zfsutils
The following NEW packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base docker.io pigz runc ubuntu-fan
0 upgraded, 8 newly installed, 0 to remove and 2 not upgraded.
Need to get 61.5 MB of archives.
After this operation, 252 MB of additional disk space will be used.
Get:1 http://ports.ubuntu.com/ubuntu-ports noble/universe arm64 pigz arm64 2.8-1 [60.7 kB]
Get:2 http://ports.ubuntu.com/ubuntu-ports noble/main arm64 bridge-utils arm64 1.7.1-1ubuntu2 [34.2
kB]
Get:3 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 runc arm64 1.3.3-0ubuntu1~24.04.
3 [8,089 kB]
Get:4 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 containerd arm64 1.7.28-0ubuntu1
~24.04.2 [27.3 MB]
Get:5 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 dns-root-data all 2024071801~ubu
ntu0.24.04.1 [5,918 B]
Get:6 http://ports.ubuntu.com/ubuntu-ports noble-updates/main arm64 dnsmasq-base arm64 2.90-2ubuntu0
.1 [366 kB]
Get:7 http://ports.ubuntu.com/ubuntu-ports noble-updates/universe arm64 docker.io arm64 28.2.2-0ubur
tu1~24.04.1 [25.5 MB]
Get:8 http://ports.ubuntu.com/ubuntu-ports noble-updates/universe arm64 ubuntu-fan all 0.12.16+24.04
.1 [34.2 kB]
Fetched 61.5 MB in 3s (22.2 MB/s)
Preconfiguring packages ...
Selecting previously unselected package pigz.
(Reading database ... 91692 files and directories currently installed.)
Preparing to unpack .../0-pigz_2.8-1_arm64.deb ...
Unpacking pigz (2.8-1) ...
Selecting previously unselected package bridge-utils.
Preparing to unpack .../1-bridge-utils_1.7.1-1ubuntu2_arm64.deb ...
Unpacking bridge-utils (1.7.1-1ubuntu2) ...
Selecting previously unselected package runc.
Preparing to unpack .../2-runc_1.3.3-0ubuntu1~24.04.3_arm64.deb ...
Unpacking runc (1.3.3-0ubuntu1~24.04.3) ...
Selecting previously unselected package containerd.
Preparing to unpack .../3-containerd_1.7.28-0ubuntu1~24.04.2_arm64.deb ...
Unpacking containerd (1.7.28-0ubuntu1~24.04.2) ...
Selecting previously unselected package dns-root-data.
Preparing to unpack .../4-dns-root-data_2024071801~ubuntu0.24.04.1_all.deb ...
Unpacking dns-root-data (2024071801~ubuntu0.24.04.1) ...
Selecting previously unselected package dnsmasq-base.
Preparing to unpack .../5-dnsmasq-base_2.90-2ubuntu0.1_arm64.deb ...
Unpacking dnsmasq-base (2.90-2ubuntu0.1) ...
Selecting previously unselected package docker.io.
Preparing to unpack .../6-docker.io_28.2.2-0ubuntu1~24.04.1_arm64.deb ...
Unpacking docker.io (28.2.2-0ubuntu1~24.04.1) ...
Selecting previously unselected package ubuntu-fan.
Preparing to unpack .../7-ubuntu-fan_0.12.16+24.04.1_all.deb ...
Unpacking ubuntu-fan (0.12.16+24.04.1) ...
Setting up dnsmasq-base (2.90-2ubuntu0.1) ...
Setting up runc (1.3.3-0ubuntu1~24.04.3) ...
Setting up dns-root-data (2024071801~ubuntu0.24.04.1) ...
Setting up bridge-utils (1.7.1-1ubuntu2) ...
Setting up pigz (2.8-1) ...
Setting up containerd (1.7.28-0ubuntu1~24.04.2) ...
Created symlink /etc/systemd/system/multi-user.target.wants/containerd.service → /usr/lib/systemd/sy
stem/containerd.service.
Setting up ubuntu-fan (0.12.16+24.04.1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/ubuntu-fan.service → /usr/lib/systemd/sy
stem/ubuntu-fan.service.
Setting up docker.io (28.2.2-0ubuntu1~24.04.1) ...
info: Selecting GID from range 100 to 999 ...
info: Adding group `docker' (GID 109) ...
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system
/docker.service.
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/doc
ker.socket.
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
verra@server1:~$
```

```
verra@server1:~$ systemctl enable docker
==== AUTHENTICATING FOR org.freedesktop.systemd1.manage-unit-files ====
Authentication is required to manage system service or unit files.
Authenticating as: Thi (verra)
Password:
==== AUTHENTICATION COMPLETE ====
verra@server1:~$ systemctl start docker
==== AUTHENTICATING FOR org.freedesktop.systemd1.manage-units ====
Authentication is required to start 'docker.service'.
Authenticating as: Thi (verra)
Password:
==== AUTHENTICATION COMPLETE ====
```

```
verra@server1:~$ systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-03-30 22:57:51 UTC; 1min 9s ago
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 1853 (dockerd)
      Tasks: 10
     Memory: 22.4M (peak: 22.9M)
        CPU: 120ms
     CGroup: /system.slice/docker.service
             └─1853 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Mar 30 22:57:50 server1 dockerd[1853]: time="2026-03-30T22:57:50.875832892Z" level=info msg="Loadin>
Mar 30 22:57:51 server1 dockerd[1853]: time="2026-03-30T22:57:51.036316817Z" level=info msg="Loadin>
Mar 30 22:57:51 server1 dockerd[1853]: time="2026-03-30T22:57:51.055671473Z" level=info msg="Docker>
Mar 30 22:57:51 server1 dockerd[1853]: time="2026-03-30T22:57:51.055705681Z" level=info msg="Initia>
Mar 30 22:57:51 server1 dockerd[1853]: time="2026-03-30T22:57:51.072416458Z" level=warning msg="CDI>
Mar 30 22:57:51 server1 dockerd[1853]: time="2026-03-30T22:57:51.072428041Z" level=warning msg="CDI>
Mar 30 22:57:51 server1 dockerd[1853]: time="2026-03-30T22:57:51.114414609Z" level=info msg="Comple>
Mar 30 22:57:51 server1 dockerd[1853]: time="2026-03-30T22:57:51.117640156Z" level=info msg="Daemon>
Mar 30 22:57:51 server1 dockerd[1853]: time="2026-03-30T22:57:51.117674156Z" level=info msg="API li>
Mar 30 22:57:51 server1 systemd[1]: Started docker.service - Docker Application Container Engine.
lines 1-22/22 (END)
```

```
verra@server1:~$ docker --version
Docker version 28.2.2, build 28.2.2-0ubuntu1~24.04.1
verra@server1:~$
```

Run Nginx Container

On server 1

```
docker run -d --name mynginx -p 8080:80 nginx
```

```
verra@server1:~$ docker run -d --name mynginx -p 8080:80 nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
f4badedbec24: Pull complete
b1a5e9a47bf9: Pull complete
0a43e039475a: Pull complete
e14d31524fc8: Pull complete
0d9786b44f2a: Pull complete
881ff00f5a79: Pull complete
5716fd893132: Pull complete
Digest: sha256:7150b3a39203cb5bee612ff4a9d18774f8c7caf6399d6e8985e97e28eb751c18
Status: Downloaded newer image for nginx:latest
a33c9af24363d69f28ef2c8fbb25f08b398b17f22aedf8018a8cea9852e34f17
verra@server1:~$ _
```

List Running Containers

```
docker ps
```

```
verra@server1:~$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS          PORTS
                         NAMES
a33c9af24363   nginx     "/docker-entrypoint.…"   23 seconds ago   Up 22 seconds   0.0.0.0:8080->80/
tcp, [::]:8080->80/tcp   mynginx
```

Check Nginx Containers

On server 1

```
curl http://192.168.1.123:8080
```

On server 2

```
curl http://192.168.1.123:8080
```

Nginx's htlm will appear

```
verra@server1:~$ curl http://10.190.139.4:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
verra@server1:~$
```

```
verra@server2:~$ curl http://10.190.139.4:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

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<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
verra@server2:~$ _
```